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Photovoltaic Minichannel Cooling Using Pulsating Ternary Hybrid Nanofluids: A Two-Phase CFD Study with Discrete Ordinates Radiation Modelling

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Abstract

Managing the temperature of photovoltaic modules is essential for limiting efficiency losses under high solar loading. This work evaluates a minichannel-based cooling system using a ternary hybrid nanofluid subjected to periodic inlet-flow modulation. Its performance is benchmarked against natural convection, water cooling, and steady mono-, binary-, and ternary-nanofluid circulation. The numerical methodology integrates three-dimensional transient CFD, discrete ordinates radiation, a multiphase mixture formulation for nanoparticle–fluid slip, and a temperature-dependent single-diode model. Simulations were conducted at a mean Reynolds number of 50, with the pulsation amplitude varied from 0.3 to 0.8 and the Strouhal number from 1 to 10. Electrical quantities were calculated using cycle-averaged PV temperatures after periodic flow behaviour was established. The best-performing pulsating case, obtained at A = 0.5 and St = 5, limited the maximum PV temperature to approximately 314.7 K. This value was about 1.6 K below that achieved with steady ternary hybrid nanofluid flow and 5 K below that obtained with water. The corresponding maximum electrical power, electrical efficiency, and exergetic efficiency were approximately 4.0 W, 22.4%, and 23.8%, respectively. Periodic acceleration and deceleration enhanced coolant redistribution, repeatedly redeveloped the thermal boundary layer, and reduced downstream heat accumulation. The principal contribution is the integrated transient evaluation of pulsating ternary hybrid nanofluid cooling within a common CFD–radiation–electrical–exergy framework. Pulsation reduced the maximum PV temperature by approximately 1.6 K relative to steady ternary hybrid nanofluid cooling, confirming an additional but incremental benefit rather than a dominant improvement over steady liquid cooling.

Keywords: photovoltaic cooling; ternary hybrid nanofluid; computational fluid dynamics; discrete ordinates radiation modelling; exergy analysis

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1. Introduction

Photovoltaic (PV) technology plays an increasingly important role in the transition towards low-carbon electricity generation. However, the conversion performance and service life of PV modules are strongly affected by their operating temperature. Power output typically decreases by approximately 0.4–0.5% for every °C rise in module temperature, while PV temperatures in hot climatic conditions may approach 70 °C, intensifying efficiency losses and material degradation [1]. This temperature increase occurs because only part of the absorbed solar radiation is converted into electricity, with much of the

remaining energy retained within the module as heat [2]. Agyekum et al. [3] also identified environmental exposure and thermal effects as important determinants of PV-system performance. Appropriate thermal-management measures are therefore required to control module temperature, preserve electrical output, and limit heat-induced deterioration.

Passive thermal-management methods are attractive because they remove heat without consuming auxiliary power. Their effectiveness, however, depends considerably on panel orientation, surrounding airflow, and heat-sink geometry. Numerical analysis of inclined PV modules under low-wind conditions revealed that the natural-convection Nusselt number changes markedly with panel inclination and wind velocity [4]. Buoyancy-driven heat removal can also be strengthened by modifying the rear-side flow path. For example, a rear enclosure that generated a chimney effect reduced the PV temperature by 17.3% [5]. When aluminium heat sinks were examined under natural and forced airflow, the forced-circulation arrangement increased power generation by 14.6% [6]. Fin geometry also affects cooling performance: rectangular fins achieved a 10.6% greater reduction in module temperature than circular fins because of their larger heat-transfer area [7]. Numerical comparison of continuous and segmented heat sinks showed that segmentation promoted vortical airflow and lowered the panel temperature by 10 °C [8]. More complex arrangements incorporating hollow fins, wavy passages, and porous inserts have also been explored; among them, the porous-layered wavy channel produced a temperature reduction of 38 °C relative to the reference module [9]. Nevertheless, the performance of passive systems remains closely linked to prevailing environmental conditions.

The dependence of passive cooling on weather conditions has encouraged the development of active systems that provide more controllable heat removal. Forced airflow, for example, was reported to decrease PV module temperature by 10–15 °C and improve electrical efficiency by as much as 15% compared with natural convection [10]. Rear-mounted air ducts have likewise produced a 12–14% enhancement in electrical efficiency relative to an uncooled module by directing air continuously across the heated surface [11]. Beyond the method of air supply, the internal geometry of the cooling passage also governs the resulting thermal performance. A combined MATLAB R 2025a and ANSYS Fluent 2024 R2 investigation of different channel arrangements found that curved-fin channels provided the greatest cooling effect, lowering the PV temperature by nearly 20 °C [12]. Although active air circulation provides more reliable cooling than natural convection, its heat-removal capacity remains constrained by the comparatively poor thermophysical properties of air.

Liquid-based systems generally provide stronger PV cooling than air-based arrangements because water possesses greater thermal conductivity and heat capacity. In one numerical comparison, water cooling decreased the module temperature by 35 °C and produced an electrical efficiency of 25%, whereas forced-air cooling achieved a temperature reduction of only 12 °C and an efficiency of 12.9% [2]. Active circulation through a rear-surface cooling loop was also reported to lower the module temperature by 25–30 °C and improve its electrical efficiency by approximately 15% [11]. Alternatively, applying a thin water film to the rear surface enhanced PV power output by approximately 10–12% [13]. Passive liquid cooling using water channels attached to the rear plate has also been examined, resulting in a module-temperature reduction of approximately 10 °C [14]. Collectively, these findings demonstrate the greater heat-removal potential of water-based cooling and provide the basis for investigating advanced coolants capable of delivering further thermal-performance improvements.

In micro- and minichannel cooling, thermal performance depends not only on passage geometry but also on the heat-transport properties of the circulating fluid. Although water offers clear advantages over air, its inherent thermal conductivity limits further enhancement under high heat loads. Nanofluids have consequently been explored as

alternative coolants because dispersed nanoparticles can modify the effective thermal properties of the base liquid. Gelis et al. [15] reported that using a SiO_2/water nanofluid improved the performance of a photovoltaic/thermal system relative to water alone. Khanjari et al. [16] compared Ag/water and Al_2O_3/water nanofluids and observed that thermal efficiency decreased as fluid velocity increased, whereas the overall efficiency benefited from nanoparticle volume fractions of approximately 8–9%. These results demonstrate that nanofluid performance depends on the combined selection of particle concentration and flow conditions. Using a mathematical model, Bouzidi et al. [17] evaluated the radiative heat-transfer behaviour of several water-based nanofluids and found that Cu nanoparticles performed better than the examined oxide particles, producing an efficiency improvement of approximately 12%. Thus, both nanoparticle composition and operating conditions must be considered when selecting a nanofluid for PV thermal management.

Ternary hybrid nanofluids have recently attracted attention because combining three nanoparticle species may provide complementary heat-transfer characteristics. In a microchannel heat sink, CuO–MgO–TiO_2/water reduced the bottom-wall temperature by as much as 3 K and decreased both thermal resistance and entropy generation at higher Reynolds numbers and particle concentrations up to 0.5 vol.%, outperforming the corresponding binary coolants [18]. A TiO_2–MgO–GO/water coolant was subsequently evaluated in minichannels of different cross-sectional shapes. It increased the heat-transfer coefficient by 17%, reduced thermal resistance by 25% and lowered the heat-sink temperature by 3 K; the trapezoidal passage achieved the highest thermo-hydraulic performance factor of 1.2 [19]. For Al_2O_3–Cu–CuO/water, a particle ratio of 30:50:20 produced a maximum Nusselt-number enhancement of 73% and reduced entropy generation by 30% relative to water [20]. Mukherjee et al. [21] reported that 0.30–0.50 wt.% CuO–MgO–GO/water enhanced the Nusselt number by up to 148%, while increasing the average friction factor by only 3% under a constant heat flux of 5.4 kW/m^2. The resulting 52% decrease in energy loss further improved its thermo-hydraulic performance. Applications beyond minichannels have shown similar benefits. Sahoo et al. [22] obtained improvements of 22% in convective heat transfer and 7% in second-law efficiency in a radiator, whereas TiO_2–Al_2O_3–ZnO/water increased the energy and exergy efficiencies of an evacuated-tube solar collector by 77% and 11%, respectively [23]. These results indicate that ternary hybrid nanofluids can improve heat transfer while limiting thermal resistance and irreversibility across different thermal systems.

Periodic modulation of coolant flow provides an additional means of intensifying convective heat transfer. Repeated acceleration and deceleration alter the near-wall velocity field, interrupt thermal-boundary-layer growth, and promote exchange between the heated wall region and the cooler fluid core. Akdag et al. [24] examined Al_2O_3/water flow through a wavy minichannel and reported a higher Nusselt number under pulsating operation than under steady flow. They further observed that pulsation amplitude had a greater effect on heat transfer than pulsation frequency. In a subsequent analysis of CuO/water flowing through a trapezoidal-corrugated channel, pulsation generated secondary-flow structures that strengthened core-to-wall fluid interaction and could also limit nanoparticle settling [25]. Akcay [26] showed that integrating pulsating nanofluid flow with semicircular corrugations and baffles intensified flow separation and thermal-boundary-layer redevelopment, although this benefit was accompanied by an increased pressure-drop requirement. Zhao et al. [27] established that periodic-flow heat transfer is controlled by the combined effects of pulsation amplitude, frequency, and nanoparticle concentration. Dawood et al. [28] further demonstrated that the velocity and temperature distributions of pulsatile nanofluid flow depend strongly on the imposed pressure waveform and channel configuration. However, these investigations predominantly concern mono-nanofluids

and non-PV channel geometries. The use of a pulsating ternary hybrid nanofluid in a PV minichannel, particularly with integrated thermal, electrical, and exergy assessment, has therefore received limited attention.

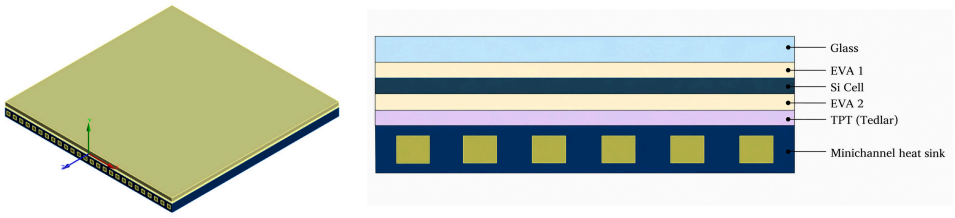
Despite extensive research on PV thermal management, three limitations remain evident in the available literature. First, most liquid-cooled PV investigations assume steady inlet conditions, leaving the potential benefit of periodic flow modulation insufficiently characterized. Second, existing pulsating-flow studies largely employ mono-nanofluids in conventional, wavy, or corrugated passages rather than ternary hybrid nanofluids in PV minichannels. Third, thermal, electrical, and exergy responses are frequently examined separately, which prevents a unified assessment of how improved heat removal influences usable PV output. The present work addresses these limitations through a transient three-dimensional CFD framework that combines a multiphase mixture formulation with discrete ordinates radiation and a temperature-dependent single-diode electrical model. Natural convection, water cooling, and steady mono-, binary-, and ternary-nanofluid flow are used as reference configurations. Pulsating ternary hybrid nanofluid flow is then examined over amplitudes of 0.3–0.8 and Strouhal numbers of 1–10 at a mean Reynolds number of 50. This approach enables the separate contributions of coolant hybridization and flow pulsation to PV temperature, electrical output, and exergetic performance to be quantified.

Unlike previous pulsating-flow investigations, which have predominantly considered mono-nanofluids in conventional, wavy, or corrugated channels [24–28], the present study examines a ternary hybrid nanofluid in a PV minichannel and evaluates its thermal, electrical, and exergetic consequences within a unified transient framework. The scientific contribution therefore lies in isolating the individual benefits of active liquid cooling, nanoparticle hybridization, and inlet-flow modulation under identical operating conditions. The results are not intended to establish pulsation as a replacement for existing hybrid nanofluid cooling strategies. Instead, they determine whether pulsation can provide a measurable additional benefit after coolant hybridization has already produced substantial thermal enhancement. The advancement is quantified relative to the internal reference configurations: at St = 5 and A = 0.5, pulsation lowers the maximum PV temperature by approximately 1.6 K relative to steady ternary hybrid nanofluid cooling and by approximately 5 K relative to water cooling. Because the configurations and operating conditions used in earlier studies differ considerably, direct numerical superiority over published systems is not claimed.

2. Methodology

2.1. Geometry Description

The investigated system comprises a 0.156 m × 0.156 m PV module thermally coupled to a rear minichannel heat sink. The heat sink contains 26 parallel channels, each having a square cross-section of 0.003 m × 0.003 m, as illustrated in Figure 1b. For the natural-convection reference case, the module is enclosed within an external-air computational domain measuring 1.1 m × 1 m × 0.55 m with an inclination of 13°. The PV laminate is represented by five solid layers arranged sequentially as glass, upper ethylene-vinyl acetate (EVA), silicon cell, lower EVA, and a Tedlar–polyester–Tedlar (TPT) backsheet. The thicknesses and thermophysical properties assigned to these layers are provided in Table 1, while the electrical specifications used in the single-diode model are listed in Table 2. Water and Al_2O_3–TiO_2–Cu/water nanofluids are circulated through the minichannels for the actively cooled cases. The properties of the base fluid and the three nanoparticle constituents are summarised in Table 3. Copper contributes high thermal conductivity, whereas the metal-oxide particles provide complementary thermal and suspension characteristics. The use of particles having different diameters may also improve particle-size distribution within the suspension and facilitate thermal transport [29].



(**a**) (**b**)

Figure 1. Computational geometry: (**a**) inclined PV module and external-air domain and (**b**) front view of the rear minichannel heat sink.

Table 1. Properties of PV layers [1].

Layer	Density (Kg/m^3)	Specific Heat C_p (KJ/Kg·K)	Thermal Conductivity (W/mK)	Thickness (mm)
Glass	2450	0.79	0.7	3.2
EVA Layer 01	960	2.09	0.311	0.5
Silicon Cell	2330	0.677	130	0.21
EVA Layer 02	960	2.09	0.311	0.5
TPT	1200	1.25	0.15	0.3

Table 2. PV panel parameters.

Parameter	Value
Type	Polycrystalline Silicon
V_{OC}	0.612 V
I_{SC}	8.221 A
V_{MPP}	0.504 V
I_{MPP}	7.718 A
η_{ref}	16–16.2%
P_{MPP}	3.89 W

Table 3. Thermophysical properties and nominal diameters of the constituent nanoparticles and base fluid [29].

Nanofluid	Density (Kg/m^3)	Specific Heat C_p (J/Kg·K)	Thermal Conductivity (W/mK)	Particle Diameter (nm)
Al_2O_3	1835.8	880	35	20
TiO_2	3016.7	711	6.4	40
Cu	5614.3	420	397	50
Water	1000	4183	0.599	-

The geometry was discretised predominantly with structured hexahedral elements. Additional refinement was applied near the PV surfaces and along the minichannel walls to resolve the steep velocity and temperature gradients in these regions, as shown in Figure 2. The first near-wall cell spacing was selected to maintain a yplus of approximately 1 in the regions where turbulence modelling was applied. Mesh sensitivity was assessed by progressively increasing the number of computational cells and monitoring the resulting PV temperature and pressure drop. A grid containing approximately 2.3 million elements was selected for the remaining simulations because further refinement produced no meaningful change in the monitored quantities (Table 4).

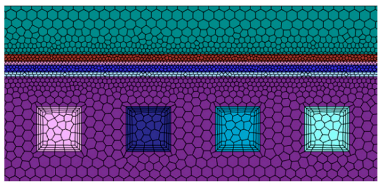


Figure 2. Computational mesh showing local refinement near the PV surface and minichannel walls.

Table 4. Grid independence study.

No. of Elements	Avg. PV T (K)	ΔP (Pa)
0.75 M	322.4	171
1.2 M	320.1	177
1.7 M	319.2	179
2.3 M (selected)	318.8	180
2.5 M	318.85	180.3

2.2. Governing Equations and Physical Models

Fluid flow, conjugate heat transfer, and radiative transport were solved using ANSYS Fluent. The external airflow surrounding the naturally cooled PV surface was represented using the unsteady Reynolds-averaged Navier–Stokes formulation with the RNG k-ε model and enhanced wall treatment [30]. Flow through the minichannels remained within the laminar regime and was therefore evaluated without a turbulence model. The governing equations were discretised using the finite-volume method. Pressure and velocity were coupled through the SIMPLE algorithm, while second-order upwind schemes were applied to the momentum and energy equations. At every time step, convergence was assessed from the scaled residuals and the stabilisation of PV temperature and channel pressure drop. The conservation equations for mass, momentum, and energy are expressed as Equations (1)–(3).

Continuity Equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \left(\rho \vec{u}\right) = 0 \tag{1}$$

Momentum Equation:

$$\rho \frac{Du}{Dt} = -\nabla p + \mu \nabla^2 \vec{u} \tag{2}$$

Energy Equation:

$$(\rho c_p)\left(\frac{\partial T}{\partial t} + (\vec{u} \cdot \nabla)T\right) = (k)\nabla^2 T \tag{3}$$

The nanofluid was represented using the mixture multiphase model rather than an effective single-phase approximation [31–33]. Water was defined as the continuous phase, while the suspended nanoparticles were introduced as dispersed secondary phases. This formulation solves common momentum and energy fields for the mixture while accounting for relative motion between the continuous and dispersed phases through an algebraic slip relation. Similar mixture-model formulations have previously been applied to nanofluid-based thermal-management systems [34].

In the mixture model, the volume fraction is computed as follows:

$$\nabla \cdot (\phi_p \rho_p \vec{V}_m) = -\nabla \cdot (\phi_p \rho_p \vec{V}_{dr,p}) \tag{4}$$

wherein, $\vec{V}_m$ is the velocity of the mixture, obtained as

$$\vec{V}_m = \sum_{s=1}^{n} \frac{\phi_s \rho_s \vec{V}_s}{\rho_m} \quad (5)$$

Drift velocity is calculated as follows:

$$\vec{V}_{dr,p} = \vec{V}_p - \vec{V}_m \quad (6)$$

Drift and slip velocity are given by the following:

$$\vec{V}_{dr,p} = \vec{V}_{pf} - \sum_{s=1}^{n} \vec{V}_{pf} \frac{\phi_p \rho_p}{\rho_m} \quad (7)$$

where $\vec{V}_{pf}$ is the slip velocity

$$\vec{V}_{pf} = \frac{\rho_p d_p^2}{18 \mu_f f_{drag}} \frac{\rho_p - \rho_m}{\rho_p} \vec{a} \quad (8)$$

The radiative interaction between the surroundings and the PV module was represented using the discrete ordinates (DO) model in combination with solar ray tracing [35]. The glass cover was specified as a semi-transparent solid so that the incident solar radiation could be partly absorbed, reflected, and transmitted through the layer [36]. The remaining solid components were treated as opaque, homogeneous, and isotropic materials. Grey radiative behaviour was assumed, and scattering within the participating medium was neglected. The surrounding air was treated as non-participating because absorption and scattering by the air were considered negligible under the investigated conditions.

The DO formulation resolves radiative intensity along a finite number of angular directions. For a participating medium, the radiative transfer equation (RTE) along direction ($\vec{s}$) can be written as follows [37]:

$$\nabla \cdot \left(I\left(\vec{r}, \vec{s}\right) \vec{s} \right) + (a + \sigma_s) I\left(\vec{r}, \vec{s}\right) = an^2 \frac{\sigma T^4}{\pi} + \frac{\sigma_s}{4\pi} \int_0^{4\pi} I\left(\vec{r}, \vec{s}'\right) \varphi\left(\vec{s}, \vec{s}'\right) d\Omega' \quad (9)$$

where r represents the position vector in space and the first term on the left-hand side represents the rate of change of radiation intensity along the direction s.

The divergence term on the left-hand side of Equation (9) describes the spatial variation in radiative intensity along the selected direction. The second term represents the removal of radiant energy from that direction through absorption and out-scattering. On the right-hand side, the first contribution accounts for thermal emission by the participating medium, whereas the integral term represents radiation redirected into direction 's' from all other directions. The phase function $\varphi\ (\vec{s}, \vec{s}')$ specifies the angular distribution of this scattering process. Because scattering was neglected in the present analysis, the out-scattering and in-scattering contributions vanish, leading to the reduced form given in Equation (10).

The scattering phenomenon is ignored, and the RTE is reduced as follows [38],

$$\nabla \cdot \left(I\left(\vec{r}, \vec{s}\right) \vec{s} \right) + (a) \left[I\left(\vec{r}, \vec{s}\right) \right] = an^2 \frac{\sigma T^4}{\pi} \quad (10)$$

The directional space in the DO formulation was divided into a finite number of solid-angle control volumes, and the radiative transfer equation was evaluated separately along each ordinate. Two angular divisions were specified in both the polar and azimuthal directions, together with a 1×1 pixelation for each angular control volume. This pixelation further subdivides the solid angle and improves the numerical evaluation of directional radiation. Solar ray tracing was coupled with the DO model to impose the solar position

and incident radiative flux corresponding to the prescribed time-dependent conditions. The total radiative intensity can be obtained by summing the intensity contributions across the defined wavelength intervals [39,40]:

$$I\left(\vec{r}, \vec{s}\right) = \sum_{k} I_{\lambda_k}\left(\vec{r}, \vec{s}\right)\Delta\lambda_k \quad (11)$$

2.3. Numerical Solution Strategy and Validation

Natural convection was employed as the reference thermal-management condition. For this case, the PV module was coupled to the surrounding air domain, and the external boundaries of the air enclosure were assigned pressure-outlet conditions to permit buoyancy-driven inflow and outflow. The ambient temperature and incident solar irradiance were varied according to the diurnal profiles presented in Figure 3a [41]. Solar radiation was applied to the semi-transparent glass cover using the solar ray-tracing model.

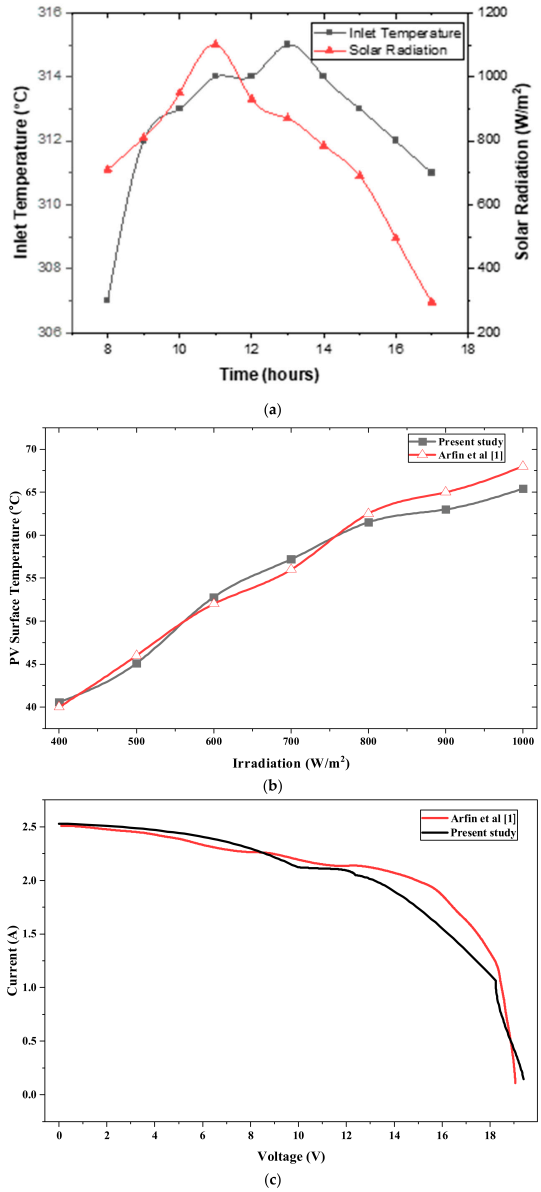


Figure 3. Model inputs and validation [1]: (**a**) diurnal variation of ambient temperature and incident solar irradiance, (**b**) comparison of predicted and experimental PV surface temperatures, and (**c**) comparison of predicted and experimental current–voltage characteristics at 1000 W/m^2.

For the liquid-cooled configurations, coolant entered the minichannels at the instantaneous ambient temperature. This boundary condition was selected to isolate the influence of coolant type and pulsation parameters under identical thermal inlet conditions. If the return temperature had been calculated without explicitly modelling the reservoir, piping, pump, and heat exchanger, it would have introduced an arbitrary thermal-recovery assumption. A velocity inlet was imposed at the channel entrance, while the outlet was maintained at zero-gauge pressure. No-slip conditions were specified at all solid–fluid interfaces. The walls separating the coolant from the PV assembly were thermally coupled to permit conjugate heat transfer, whereas only the externally insulated surfaces of the minichannel heat sink were treated as adiabatic.

The total nanoparticle volume fraction was varied between 1 and 3 vol.%. For the ternary hybrid nanofluid, Al_2O_3, TiO_2, and Cu were combined in unequal 40:40:20 (unequal) and 1:1:1 (equal) proportions. The 40:40:20 composition was adopted based on previous experimental findings showing that this ratio provides a suitable balance between thermal-conductivity enhancement and increased viscosity [29]. The 1:1:1 composition was evaluated to determine the effect of nanoparticle distribution at an identical total particle loading. Convective heat exchange between the exposed PV surface and ambient air was calculated using the wind-dependent correlation

$$h = 3.6w + 2.92 \tag{12}$$

where w is the wind velocity and is taken as 1.5 m/s.

At the inlet, the velocity profile is found by adding the uniform velocity profile which is 0.016 m/s with a sinusoidal pulsation. Thus, the inlet velocity profile is given by the following:

$$u_{(in)} = u_m(1 + A\sin(2\pi ft)) \tag{13}$$

where A is the non-dimensional amplitude and f is the frequency defined as follows:

$$f = Stu_m/H_d \tag{14}$$

where St is non dimensional Strouhal number, U_m is the mean velocity, and H_d is the hydraulic diameter.

The CFD-derived PV surface temperature, exported at 60 s intervals, is provided as input to a MATLAB single-diode electrical model:

$$I = I_{ph} - I_0\left[\exp\left(\frac{q(V + IR_s)}{nkT}\right) - 1\right] - \frac{V + IR_s}{R_{sh}} \tag{15}$$

where I_{ph} is the photocurrent, I_0 is the saturation current, R_s and R_{sh} are the series and shunt resistances, n is the ideality factor, and T is the CFD-predicted cell temperature.

The thermal and electrical components of the numerical framework were assessed independently using the experimental measurements reported by Arifin et al. [1]. For thermal validation, the geometry, material properties, and operating conditions of the reference experiment were reproduced in the CFD model. Figure 3b compares the predicted and measured PV surface temperatures over the investigated solar-irradiance range. The numerical temperatures follow the experimental trend, with an average deviation of 3.04%.

The electrical model was evaluated at an irradiance of 1000 W/m^2 by comparing the predicted current–voltage curve with the corresponding experimental data, as shown in Figure 3c. The maximum deviations in voltage and current were 8.1% and 9.5%, respectively. In addition, the implementation of the multiphase mixture formulation was previously compared with experimental nanofluid heat-transfer data in the authors' earlier

work [33]. The agreement obtained in these separate assessments supports the use of the coupled framework for the present thermo-electrical analysis. A sequential one-way thermo-electrical coupling was adopted. The CFD-predicted cell temperature was supplied to the MATLAB single-diode model to determine temperature-dependent electrical output. The electrical solution was not subsequently returned to the CFD domain as an updated spatial heat-generation field. The procedure therefore captures the first-order effect of temperature on electrical performance but neglects the secondary thermal feedback associated with temperature-dependent electrical and ohmic losses. A strongly coupled solution would require iterative exchange between the CFD and electrical solvers at every time step until the temperature, electrical power, and internal heat-generation fields converge. The present electrical results should consequently be interpreted within the one-way coupling assumption.

2.4. Data Reduction

The following equations are used to derive various parameters.
The Pumping power is given by:

$$P_m = \frac{\dot{m}}{\rho}\Delta P \tag{16}$$

where ΔP is the Pressure Drop in the channel, and $\dot{m}$ is the mass flow rate given by

$$\dot{m} = \rho A V \tag{17}$$

The Power of the PV is given by the following:

$$P_{elec} = V * I \tag{18}$$

The Net Power Gain can be expressed as

$$P_{net\ gain} = P_{elec} - P_m \tag{19}$$

The Electrical Efficiency of the PV Panel is determined by the following [42]:

$$\eta_{ele}(\%) = (P_{net\ gain}/(A_{pv} \times G_{pv})) \times 100 \tag{20}$$

The electrical exergy efficiency can be expressed as follows:

$$\varepsilon_{el} = \frac{\dot{E}x_{el}}{\dot{E}x_{sun}} = \frac{\dot{E_{el}} - \dot{E_P}}{\dot{G} \cdot A_c \cdot \left(1 - \frac{T_{amb}}{T_{sun}}\right)} = \frac{V_{oc} \cdot I_{sc} - \dot{E_P}}{\dot{G} \cdot A_c \cdot \left(1 - \frac{T_{amb}}{T_{sun}}\right)} \tag{21}$$

The Thermal Exergetic efficiency is expressed as follows:

$$\varepsilon_{th} = \frac{\dot{E}x_{th}}{\dot{E}x_{sun}} = \frac{\dot{m} \cdot C_{p,f}\left[(T_{f,out} - T_{f,in}) - T_{amb}\ln\left(\frac{T_{f,out}}{T_{f,in}}\right)\right]}{\dot{G} \cdot A_c \cdot \left(1 - \frac{T_{amb}}{T_{sun}}\right)} \tag{22}$$

Therefore, the Overall Exergetic efficiency can be calculated as follows:

$$\varepsilon_{ov} = \varepsilon_{el} + \varepsilon_{th} \tag{23}$$

3. Results and Discussion

The thermo-electrical and exergetic performances of the photovoltaic module are compared under six thermal-management configurations: natural convection without active cooling, water cooling, steady mono-nanofluid cooling, steady binary hybrid nanofluid cooling, steady ternary hybrid nanofluid cooling, and pulsating ternary hybrid nanofluid cooling. All actively cooled cases are evaluated under an identical mean Reynolds number and the same peak solar irradiance and ambient conditions, thereby isolating the effects of coolant composition and inlet-flow pulsation. The discussion first examines the resulting PV surface temperature and electrical efficiency, followed by an evaluation of the current–voltage and power–voltage characteristics. Finally, the exergetic efficiencies are compared to determine whether the improvements in heat removal and electrical power are accompanied by better utilisation of the available solar exergy.

3.1. Thermo-Electrical Performance

3.1.1. Diurnal Variation in PV Temperature and Electrical Efficiency

The diurnal variation in PV surface temperature and electrical efficiency under natural convection and water cooling are as shown in Figure 4. For the uncooled PV module, the surface temperature increases from approximately 365 K to a maximum of nearly 369 K around midday. This temperature rise follows the increase in incident solar irradiance, as a major fraction of the absorbed solar energy is dissipated as heat. Under natural convection, heat rejection is restricted to buoyancy-driven airflow and surface radiation; therefore, the rate of heat removal is insufficient to prevent heat accumulation during periods of high irradiance. The elevated operating temperature negatively affects the electrical performance of the PV module. As shown in Figure 4, the electrical efficiency of the uncooled configuration remains lower than that of the water-cooled configuration throughout most of the daytime operating period. Increasing cell temperature raises the semiconductor saturation current and reduces the open-circuit voltage. Although the short-circuit current may increase marginally with temperature, this improvement is insufficient to compensate for the voltage loss. Consequently, the electrical conversion efficiency decreases as thermal loading increases.

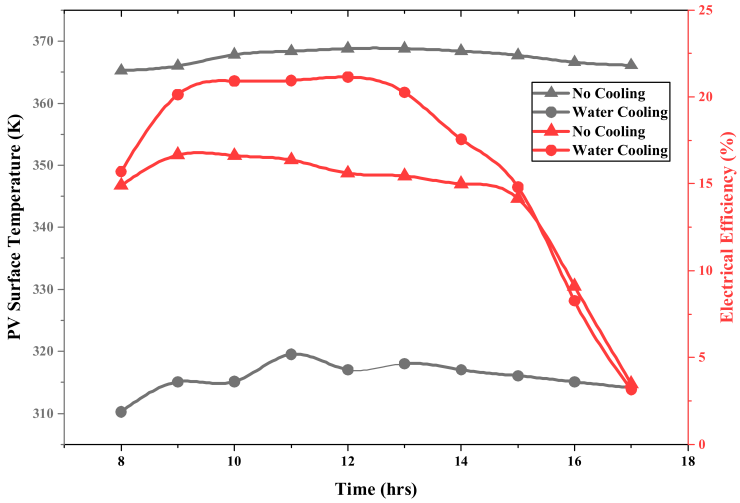


Figure 4. Diurnal evolution of PV surface temperature and electrical efficiency under natural-convection and water-cooled operating conditions.

The introduction of water cooling substantially alters the diurnal thermal response. The water-cooled PV temperature remains within approximately 310–320 K, representing a reduction of nearly 50 K relative to natural convection during the high-irradiance period

(Figure 4). The lower temperature is attributed to the greater specific heat capacity and convective heat-removal capability of water. The improved thermal regulation is directly reflected in the electrical efficiency. Water cooling maintains an electrical efficiency of approximately 20%, whereas the corresponding value for the naturally cooled module is approximately 15%. By limiting the rise in junction temperature, water cooling suppresses thermally induced voltage degradation and preserves a higher maximum power output.

3.1.2. Comparative Performance of Steady and Pulsating Coolant Flow

As water cooling proved to be superior, PV surface temperature and electrical efficiency for the uncooled, water-cooled, steady nanofluid-cooled, and pulsating nanofluid-cooled configurations under peak solar irradiance are studied as shown in Figure 5. The uncooled configuration exhibits the highest PV temperature and the lowest electrical efficiency, whereas all liquid-cooled configurations provide substantial thermal and electrical improvements. This comparison confirms that forced liquid circulation is necessary to maintain the PV module within a favourable operating-temperature range under high thermal loading.

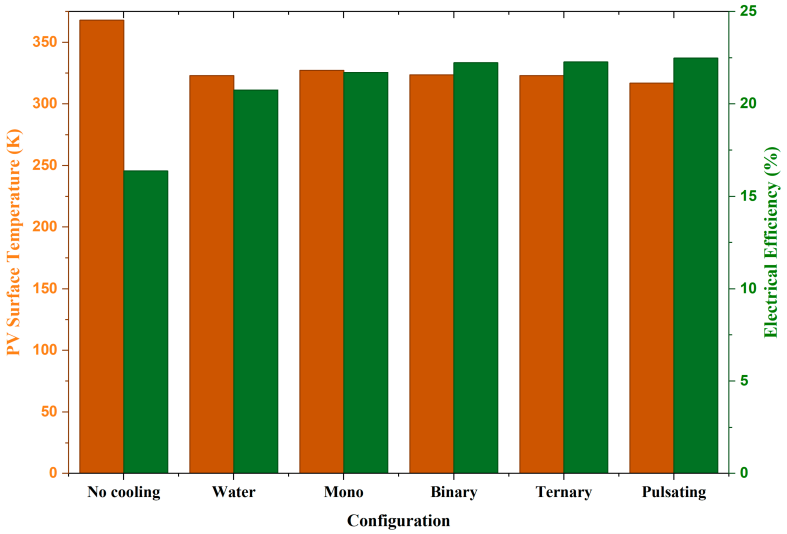


Figure 5. Comparison of PV surface temperature and electrical efficiency under different steady and pulsating cooling configurations at peak solar irradiance.

Water cooling substantially reduces the PV surface temperature compared with natural convection. Nevertheless, replacing water with nanofluids produces an additional improvement because the suspended nanoparticles increase the effective thermal conductivity of the coolant. The enhanced energy transport within the fluid facilitates heat transfer from the heated minichannel wall to the coolant core. The improvement is reflected in the corresponding increase in electrical efficiency shown in Figure 5.

Among the steady nanofluid cases, the mono-nanofluid provides the smallest improvement, while the binary and ternary hybrid nanofluids exhibit better thermo-electrical performance. The enhancement obtained through hybridisation can be attributed to the complementary thermophysical characteristics of the constituent nanoparticles. Combining nanoparticles with different thermal conductivities, densities, and specific heat capacities can provide a more favourable balance between thermal conductivity and coolant transport properties than a single-particle nanofluid. However, the difference between the binary and ternary configurations is relatively small, indicating diminishing thermal benefits with successive nanoparticle hybridisation. The steady ternary hybrid nanofluid produces a lower PV temperature and higher electrical efficiency than water and the mono-nanofluid.

Its improved performance results from the combined contributions of Al_2O_3, TiO_2, and Cu nanoparticles.

Further, the effect of pulsating flow is studied for ternary nanofluid for amplitude and frequency varied in the range of 0.3–0.8 and 1–10, respectively. The PV surface temperature decreases to approximately 316 K, corresponding to reductions of approximately 5 K and 2 K relative to water cooling and steady ternary hybrid nanofluid cooling, respectively. The best performance is obtained using pulsating ternary hybrid nanofluid flow at St = 5 and A = 0.5. The corresponding electrical efficiency reaches approximately 22.4%, which is the highest value among all the investigated configurations. The additional improvement produced by pulsation is associated with the periodic variation in coolant velocity. During the acceleration phase, the increased instantaneous velocity strengthens convective transport and reduces the thickness of the thermal boundary layer. During the deceleration phase, the altered velocity gradient promotes redistribution between the heated near-wall fluid and the cooler channel-core fluid. Repetition of this process restricts continuous thermal-boundary-layer development and improves the cycle-averaged heat-transfer rate compared with steady flow at the same mean Reynolds number. However, the incremental improvement over steady ternary-nanofluid cooling is smaller than the improvement achieved by introducing water cooling. This indicates that active liquid cooling provides the dominant thermal benefit, while nanoparticle hybridisation and pulsation deliver successive refinements.

3.1.3. Current–Voltage and Power–Voltage Characteristics

The influence of the cooling configuration on the electrical behaviour of the PV module is further examined using the current–voltage characteristics as shown in Figure 6. The uncooled configuration exhibits the lowest open-circuit voltage and an earlier decline in current as the terminal voltage increases. This behaviour is associated with its elevated cell temperature, which increases the reverse saturation current of the diode and reduces the voltage developed across the PV junction. Water cooling shifts the knee of the I–V curve towards a higher voltage (Figure 6). The reduction in PV temperature limits the temperature-induced deterioration of the open-circuit voltage and allows the module to maintain a higher current over a wider voltage range. Consequently, water cooling improves both the operating voltage and the fill factor relative to the uncooled configuration.

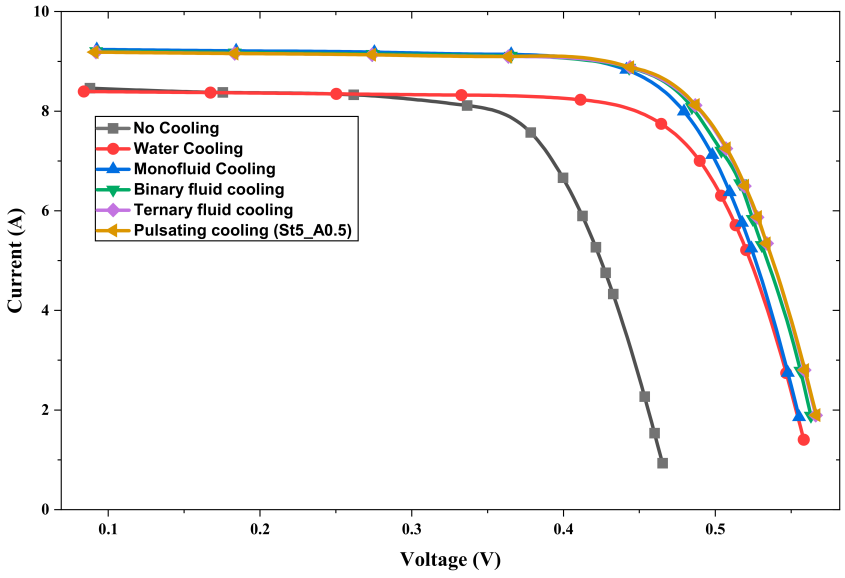


Figure 6. Current–voltage characteristics of the PV module under different steady and pulsating cooling configurations at peak solar irradiance.

The mono-, binary-, and ternary-nanofluid cases exhibit a further rightward shift of the I–V curve. The improvement becomes progressively more pronounced from mono- to binary- and ternary-nanofluid cooling, consistent with the reduction in PV temperature reported in Figure 5. The differences among the nanofluid curves are relatively small in the low-voltage region but become more apparent near the knee and open-circuit conditions. This indicates that the principal electrical advantage of improved cooling arises from voltage preservation rather than a substantial change in the short-circuit current.

The pulsating ternary hybrid nanofluid produces a favourable I–V characteristic similar to that of ternary nanofluid. At St = 5 and A = 0.5, its curve maintains a higher current close to the maximum-power region and exhibits the highest open-circuit voltage among the investigated configurations.

The effect of these changes on electrical power is evident from the power–voltage curves presented in Figure 7. The uncooled PV module produces the lowest maximum power, approximately 3 W, because of its reduced operating voltage and less distinct knee region. Although the current remains comparatively stable at low voltage, the early voltage-induced decline in current restricts the maximum obtainable power. Water cooling increases the maximum power to approximately 3.6 W (Figure 7). The corresponding power peak also shifts towards a higher operating voltage, demonstrating that the thermal benefit of water cooling is translated directly into improved electrical output. This increase is mainly associated with recovery of the operating voltage and improvement in the fill factor.

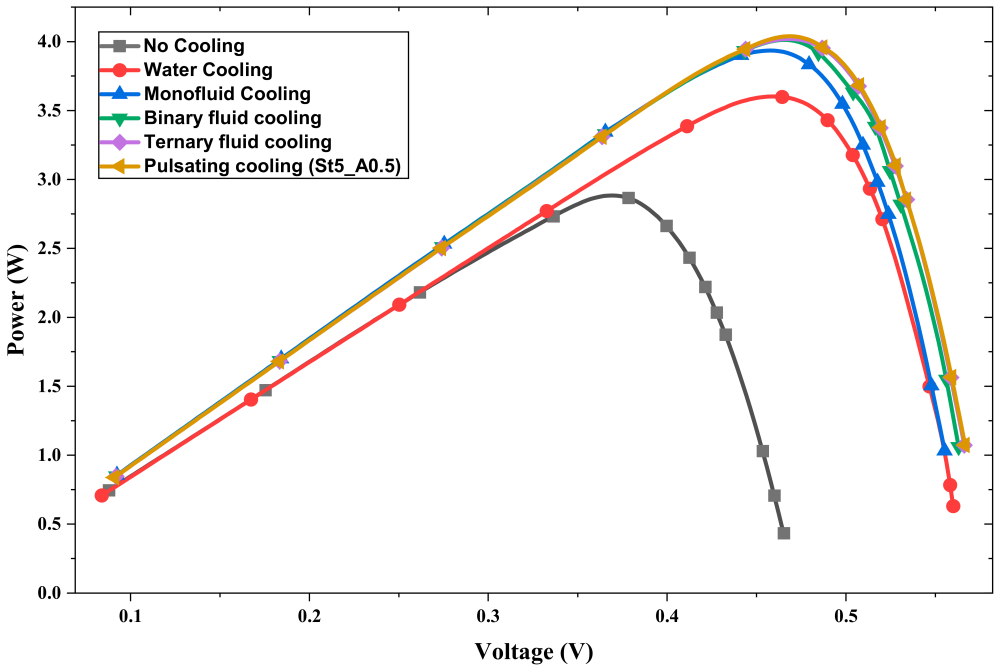


Figure 7. Power–voltage characteristics of the PV module under different steady and pulsating cooling configurations at peak solar irradiance.

The nanofluid-cooled configurations produce maximum-power values of approximately 4 W. The mono-nanofluid provides a moderate improvement over water, whereas the binary and ternary hybrid nanofluids generate slightly higher and broader power peaks. The relatively small separation between the binary and ternary curves indicates that the incremental electrical benefit of adding a third nanoparticle is limited once effective liquid cooling has already reduced the cell temperature substantially. The pulsating ternary hybrid nanofluid produces the maximum power, similar to ternary nanofluid, approximately 4 W, at St = 5 and A = 0.5 (Figure 7). Its power peak occurs at a higher voltage than those of the uncooled, water-cooled, and mono-nanofluid-cooled configurations. This improve-

ment confirms that the lower PV temperature obtained under pulsating flow enhances the usable electrical output rather than merely altering the thermal state of the module. The separation between the steady ternary and pulsating ternary power curves is modest. Therefore, pulsation should be interpreted as an incremental enhancement to an already effective ternary-nanofluid cooling system. Its practical benefit must ultimately be assessed against the additional pumping and flow-control requirements associated with producing the oscillatory inlet condition. The progressive shift of the I–V knee and P–V peak from the uncooled configuration towards the pulsating ternary hybrid nanofluid configuration is consistent with the temperature and efficiency trends presented in Figure 5.

The spatial distribution of PV surface temperature under water and steady ternary hybrid nanofluid cooling is presented in Figure 8 for Reynolds numbers of 50. These contours provide qualitative insight into the thermal behaviour underlying the electrical-performance trends discussed in Figures 5–7. Under water cooling, a pronounced temperature gradient is observed with the PV surface temperature increasing progressively towards the downstream region. Fresh coolant entering the minichannels initially provides a relatively high temperature difference for heat transfer. As the water flows downstream, it absorbs heat and its local temperature increases, reducing the temperature difference between the PV surface and the coolant. Consequently, heat removal becomes less effective near the outlet, producing a downstream high-temperature region.

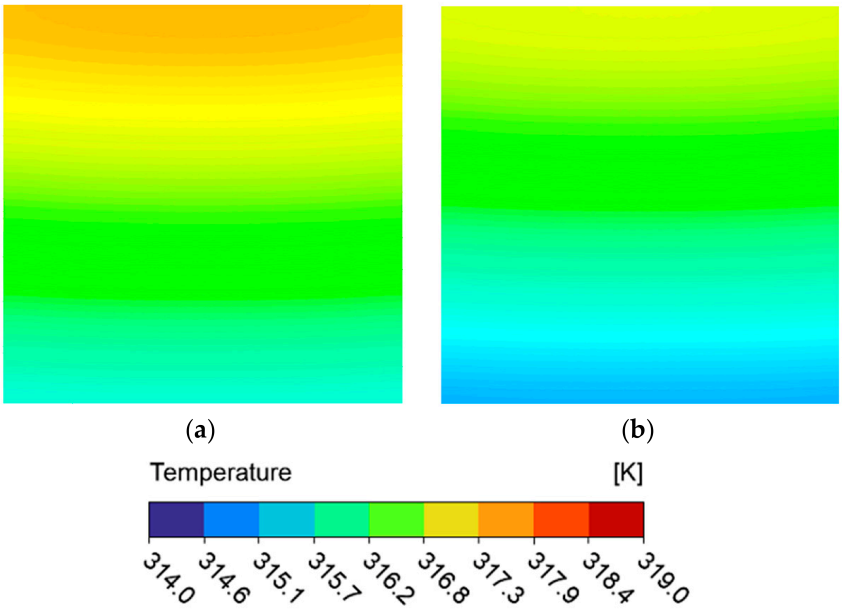


Figure 8. PV surface-temperature contours obtained with (**a**) water and (**b**) ternary hybrid nanofluid cooling.

The ternary hybrid nanofluid produces a lower and more uniform PV surface temperature than water at the corresponding Reynolds numbers. The improvement is particularly evident where the downstream high-temperature region is substantially suppressed. The enhanced effective thermal conductivity and thermal diffusivity of the ternary hybrid nanofluid facilitate energy transport away from the solid–fluid interface and improve heat redistribution within the coolant.

The improved temperature uniformity is electrically important because localised high-temperature regions can produce non-uniform cell voltage and increase mismatch losses. Therefore, the contours in Figure 8 support the improved I–V and P–V characteristics obtained using ternary hybrid nanofluid cooling. They also establish the steady ternary

configuration as the appropriate reference against which the additional benefit of pulsating flow is evaluated.

3.2. Exergetic Performance

The overall exergetic efficiency of the PV system under the different cooling configurations is compared in Figure 9. Unlike energy efficiency, exergy efficiency accounts for both the quantity and quality of the recovered electrical and thermal outputs relative to the exergy content of the incident solar radiation. It therefore provides a more rigorous measure of the effectiveness with which the available solar-energy potential is utilised. The uncooled PV module exhibits the lowest exergetic efficiency, approximately 17.2%. Under natural convection, the elevated PV temperature reduces the electrical power output through voltage degradation. Moreover, because no circulating coolant is employed, there is no recoverable thermal exergy associated with a useful coolant-temperature rise. These factors result in poor utilisation of the incident solar exergy. Water cooling increases the overall exergetic efficiency to approximately 21.9% (Figure 9). The reduction in PV temperature improves the electrical exergy component by restoring the operating voltage and maximum power. Simultaneously, the temperature rise of the circulating water provides a thermal exergy contribution. Although the thermal energy absorbed by the coolant may be substantial, its exergy content remains comparatively limited because the coolant outlet temperature is relatively close to the ambient reference temperature.

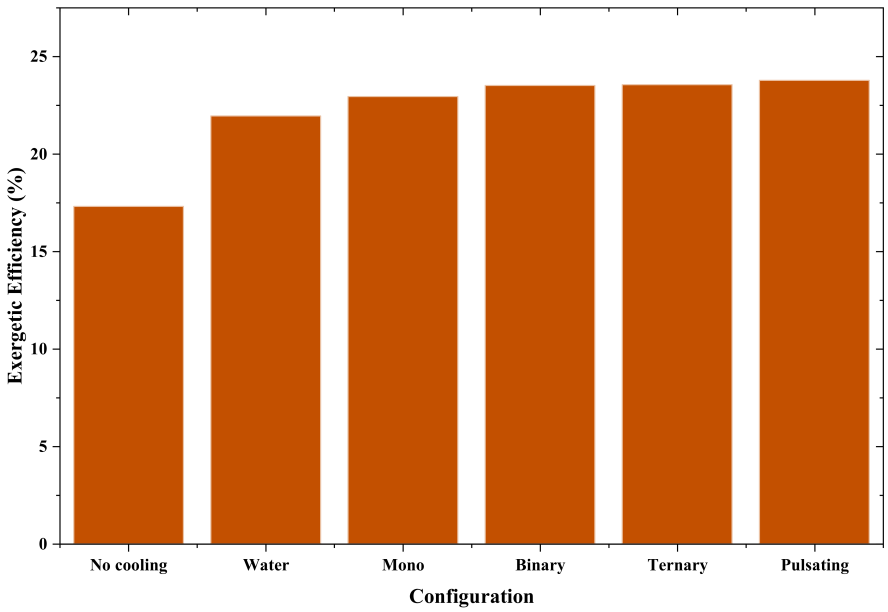


Figure 9. Overall exergetic efficiency of the PV system under different steady and pulsating cooling configurations at peak solar irradiance.

The mono-nanofluid produces an exergetic efficiency of approximately 22.9%, representing a moderate improvement over water cooling. This enhancement follows the improved thermo-electrical response observed in Figures 5–7. The binary hybrid nanofluid further increases the exergetic efficiency to approximately 23.5%, while the steady ternary hybrid nanofluid delivers similar results. The progressively smaller improvement from mono- to binary- and ternary-nanofluid cooling indicates diminishing exergetic returns from successive coolant hybridisation.

The pulsating ternary hybrid nanofluid exhibits overall exergetic efficiency similar to ternary nanofluid, approximately 23.8%, at St = 5 and A = 0.5 (Figure 9) under peak load condition. Compared with the uncooled configuration, pulsating cooling increases the

exergetic efficiency by approximately 6.6 percentage points. The improvement over water cooling is approximately 1.9 percentage points, whereas the increase relative to steady ternary hybrid nanofluid cooling is not significant.

Overall, the exergy trend in Figure 8 is consistent with the thermal and electrical results. The transition from natural convection to water cooling provides the largest improvement, while mono-, binary- and ternary-nanofluids provide successive but smaller gains. It is observed that binary and ternary nanofluids exhibit similar exergetic efficiency.

4. Conclusions

The present study numerically investigated the thermal, electrical, and exergetic performance of a minichannel-cooled photovoltaic module under natural convection, water cooling, steady mono-, binary-, and ternary-nanofluid cooling, and pulsating ternary hybrid nanofluid cooling. A coupled three-dimensional CFD–electrical framework was employed using the discrete ordinates radiation model, a multiphase mixture model, and a temperature-dependent single-diode electrical model. The effects of coolant hybridisation and inlet-flow pulsation were evaluated under identical irradiance and ambient conditions. The principal conclusions are as follows:

- Under natural convection, the PV surface temperature reached approximately 369 K around midday, while its electrical efficiency remained approximately 15%. Water cooling maintained the PV temperature within approximately 310–320 K and increased the electrical efficiency to approximately 20%, demonstrating that active liquid cooling provides the dominant thermal-management benefit.
- Mono-nanofluid cooling provided a moderate improvement over water, whereas the binary and ternary hybrid nanofluids achieved lower PV temperatures and higher electrical efficiencies. However, the comparatively small difference between the binary and ternary configurations indicated diminishing performance enhancement with successive nanoparticle hybridisation.
- The selected pulsating ternary hybrid nanofluid condition at St = 5 and A = 0.5 provided the best thermo-electrical performance. The PV surface temperature decreased to approximately 316 K, corresponding to reductions of approximately 5 K and 2 K relative to water and steady ternary hybrid nanofluid cooling, respectively. The corresponding electrical efficiency reached approximately 22.4%.
- The PV surface-temperature contours demonstrated that steady ternary hybrid nanofluid cooling suppressed the downstream high-temperature region and produced a more uniform temperature distribution than water cooling. This improved thermal uniformity can reduce localised voltage degradation and thermally induced mismatch across the PV surface.
- Improved cooling shifted the knee of the I–V curve and the peak of the P–V curve towards higher operating voltages. The maximum power increased from approximately 3 W without active cooling to 3.6 W with water cooling and approximately 4 W with ternary and pulsating ternary hybrid nanofluid cooling. The relatively small separation between the steady and pulsating ternary cases indicated that pulsation provides an incremental rather than a dominant electrical benefit.
- The overall exergetic efficiency increased from approximately 17.2% for the uncooled module to 21.9% with water cooling, 22.9% with mono-nanofluid cooling, and approximately 23.5–23.8% with binary, ternary, and pulsating ternary hybrid nanofluid cooling. Pulsating flow produced the highest value of approximately 23.8%; however, its improvement over steady ternary-nanofluid cooling was not significant.

Overall, the results demonstrate that active liquid cooling produces the greatest improvement in PV performance, while nanoparticle hybridisation and inlet-flow pulsation provide successive refinements. Pulsating ternary hybrid nanofluid flow achieved the lowest PV temperature and highest electrical and exergetic efficiencies, but its modest advantage over steady ternary-nanofluid flow must be weighed against the additional pumping, control, and operational requirements associated with generating oscillatory flow. Because the investigated pulsation cases did not constitute a sufficiently dense two-dimensional factorial design, continuous response contours of maximum PV temperature and average Nusselt number were not constructed. Accordingly, the joint interaction between St and A, as well as the local or global nature of the selected condition, cannot be established from the present dataset.

The present study has adopted the following assumptions in the numerical framework. The pulsating inlet condition was prescribed using an ideal sinusoidal velocity waveform and therefore does not account for deviations associated with practical pulsating pumps. Long-term nanoparticle agglomeration, sedimentation, and deposition or changes in particle distribution are not accounted for in the mixture model. In addition, the electro-thermal coupling is one-way: the CFD-predicted PV temperature is supplied to the temperature-dependent single-diode electrical model, while electrical heat generation is not fed back into the CFD solution. For the liquid-cooled cases, the coolant inlet temperature was prescribed at the instantaneous ambient temperature, and its associated temperature rise that would occur in a closed loop condition is not accounted for in the simulation. Consequently, the reported thermal, electrical, and exergetic improvements represent the behaviour of the idealized computational system under the investigated operating conditions and should not be interpreted as direct predictions of long-term practical performance. The present results consequently support the robustness of the selected condition under the examined day, while a comprehensive time-dependent optimization remains necessary before implementing an adaptive pulsation-control strategy.

Future studies should quantify cycle-averaged pressure drop and pumping power across a wider range of Reynolds numbers, amplitudes, Strouhal numbers, and pulsation waveforms. Long-term nanoparticle stability, agglomeration, sedimentation, erosion, and possible minichannel blockage should also be investigated before practical implementation. Addressing these aspects could facilitate the development of efficient and reliable PV thermal-management systems, contributing to SDG 7 (Affordable and Clean Energy), SDG 9 (Industry, Innovation and Infrastructure), and SDG 13 (Climate Action).

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Nomenclature

Symbol	Description	Unit
A	Area	m^2
c_p	Specific heat capacity	J kg^{-1} K^{-1}
C_d	Drag coefficient	–
d_p	Particle/droplet diameter	m
G	Solar irradiance	W m^{-2}
h	Convective heat transfer coefficient	W m^{-2} K^{-1}
I	Radiation intensity	W m^{-2}
I_{sc}	Short-circuit current	A
k	Thermal conductivity	W m^{-1} K^{-1}
Kn	Knudsen number	–
$\dot{m}$	Mass flow rate	kg s^{-1}
Nu	Nusselt number	–
P_{MPP}	Maximum power point	W
ΔP	Pressure drop	Pa
Re	Reynolds number	–
Sh	Sherwood number	–
T	Temperature	K
u	Fluid velocity	m s^{-1}
u_p	Particle velocity	m s^{-1}
V_{oc}	Open-circuit voltage	V
η_{el}	Electrical efficiency	–
η_{ex}	Exergetic efficiency	–
μ	Dynamic viscosity	Pa s
ρ	Density	kg m^{-3}
σ	Stefan–Boltzmann constant	W m^{-2} K^{-4}
Acronym	Description	
CFD	Computational Fluid Dynamics	
PV	Photovoltaic	
DO	Discrete Ordinates	
DPM	Discrete Phase Model	
RTE	Radiative Transfer Equation	
Symbol	Description	
ex	Exergy	
f	Fluid	
in	Inlet	
out	Outlet	
p	Particle/droplet	
pv	Photovoltaic cell	

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